

Expert Value Table: Accuracy vs. Distance

Two different real-world contexts appear in the research, so I split the table into them — they show the same curve shape (high accuracy close up, dropping off faster as distance increases) but very different starting points depending on training level and stress.

A. Real-world / stress conditions (pistol engagements)

Source: [NYPD SOP-9 statistics](#), [Officer-Involved Shootings report](#)

Distance	Expected Hit Rate
Under 3 m	~45–55%
3–7 m	~10–15%
7–15 m	~4–8%
15–25 m	~3–6%
25+ m	~2–4%

B. Trained / controlled conditions (rifle, stable firing position)

Source: [Effect of firing position on aiming error and probability of hit](#)

Distance	Expected Hit Probability	Notes
~100 m	~90–99%	Stable position, minimal error
~200 m	~80–90%	Small aiming errors start to matter more
~300 m	~65–99% (position-dependent)	Prone: ~99% • Kneeling: ~68% — position stability becomes a major factor here

C. Precision/group-size scaling (how "spread" grows with distance)

Source: [Precision & Group Size measurement guide](#), [Mean Radius explained](#)

Distance	Expert group spread (≈1 MOA shooter)
100 m	~2–3 cm
200 m	~4–6 cm
300 m	~6–9 cm

Speed vs. Parameters

1. Distance / Target Size (Fitts's Law)

Source: [Speed-accuracy tradeoff in Fitts' law tasks](#)

Condition	Expert engagement speed
Close range, large target	Fastest — near minimum possible time
Far range / small target	Progressively slower, but each further increase adds proportionally less extra time (logarithmic, not linear)

2. Number of Choices / Targets to Decide Between (Hick's Law)

Source: Hick's Law — [Wikipedia](#), [Hick's Law for Choice Reaction Time: A Review](#)

Number of options	Typical decision time
1 (no real choice)	~260 ms
2	~320–330 ms
4	~340–350 ms
8	~450–460 ms

With practice, the gap between few-choice and many-choice situations shrinks substantially (one study found the 2-vs-8-choice gap fall from ~500 ms to ~300 ms after repeated trials) — meaning experienced trainees should show a flatter "choices vs. time" curve than novices.

3. Fatigue

Source: [Acute fatigue affects reaction times in MMA fighters](#)

State	Effect on speed
Fresh	Baseline reaction time
Fatigued	~1–2% slower on average, and notably less consistent (~15% more variable) — fatigue hurts reliability more than raw speed

4. Stress / Arousal Level (Yerkes-Dodson Law)

Source: [Yerkes-Dodson Law of Arousal and Performance](#)

Arousal level	Effect on speed
Low (bored/under-stimulated)	Slower than optimal

Optimal (moderate stress)	Fastest, most accurate
High (panic/overload)	Slower again – worse on complex tasks than simple ones

5. Experience / Training Level

Source: [Domain-specific reaction times: experienced vs. novice goalkeepers](#)

Skill level	Effect on speed
Novice	Slower, especially as choices/targets increase
Expert	Faster, but <i>only</i> on tasks matching their trained domain – the advantage doesn't carry over to unrelated tasks

Combined view

Parameter	Low end (slow)	High end (fast/expert)
Distance/target size	Far & small	Close & large
Number of choices	8+ options (~450ms)	1–2 options (~260–330ms)
Fatigue	Fatigued (+1–2% slower, less consistent)	Fresh
Arousal	Very low or very high	Moderate/optimal
Experience	Novice	Domain-expert

Safety Score — Expert Values for Each Piece

Piece A — Outcome (rescue success rate)

Source: RAND — [hostage rescue attempts](#), [Crisis negotiation success rates](#)

Resolution approach	Hostage survival rate
Negotiated / controlled resolution	~85–90% resolved without violence
Tactical rescue assault	~70–75% survive (~25–31% killed)
No intervention / abandoned	<40% survive (60%+ killed by captors)
Overall real-world average (all methods)	~80% survive

This gives you a real benchmark for "Outcome = 1": in the real world, even the best approach (negotiation) doesn't hit 100% — so treating rescue success as binary (1 or 0) is reasonable, but know that ~85-90% is roughly the ceiling professionals achieve, not 100%.

Piece B — Well-Being (distress states)

Source: [Polyvagal Theory — fight/flight/freeze/fawn](#), [Subjective Units of Distress Scale \(SUDS\)](#)

Your five states map almost exactly onto the clinically-recognized trauma-response stages:

Your state	Matches (Polyvagal/trauma research)	Distress meaning
Calm	Ventral vagal (safe, socially engaged)	Baseline safety
Follow	Fawn (appeasement/submission to threat)	Low but present stress
Fearful	Sympathetic activation (fight/flight)	Moderate-high stress
Freeze	Dorsal vagal shutdown	High stress, immobilized
Panic	Extreme sympathetic dysregulation	Peak distress

The standard clinical tool for quantifying distress (SUDS) uses the same idea as your percentages — a 0–100 self-rated distress scale, where 0 = no distress, 100 = worst distress ever experienced. Your 0/10/33/50/66 values are a reasonable compressed version of that same scale, and the ordering (Calm → Follow → Fearful → Freeze → Panic) matches the accepted escalation sequence in the research, so the shape of Piece B is well-grounded even though the exact percentages are your own calibration.

Piece C — Friendly Fire penalty

Source: [Friendly Fire: Facts, Myths and Misperceptions](#), [Friendly fire trends since Desert Storm](#)

Context	Real-world friendly fire rate
Vietnam War	~5% of total casualties
Gulf War (Desert Storm)	~17–23% fratricide rate
Modern trained units	<2% (down ~40% since Desert Storm improvements)

This supports treating any friendly-fire incident as a serious red flag rather than a minor deduction — real trained units aim for near-zero, so your steep first-incident penalty (39%) is well aligned with how seriously this is treated operationally.

One more thing worth flagging: your penalty curve (0%, 39%, 63%, 78%) fits almost exactly the pattern $1 - (0.61)^n$ — a standard diminishing-returns/saturating decay shape. That's the same mathematical family used in reliability and cumulative-risk modeling (each additional bad event matters less than the last because the outcome is already heavily damaged) — so mathematically your curve is internally consistent,

even though I didn't find a study that derives those exact percentages for friendly fire specifically.

Combined reference table

Piece	Expert/best-case value	Worst-case value
Outcome (rescue)	~85–90% success achievable	<40% if mishandled
Well-Being	Mostly Calm/Follow states	Extended Freeze/Panic
Friendly Fire	Near 0% (modern trained units)	17–23%+ (historical poorly-coordinated ops)

Expert Value Table: Player Safety Behavior

Parameter	Novice / Untrained	Expert / Trained	Source
Shoot/no-shoot false positives (shooting a non-threat)	Noticeably higher — cadets show "a considerable number" of false positives in field studies	Lower, but never fully eliminated — persists under high-anxiety scenarios even after training	Blended firearms training study , Developing scenarios that evoke shoot/don't-shoot errors
Shoot/no-shoot false negatives (failing to engage a real threat)	Present at baseline	Reduced to near-zero after targeted training	Blended police firearms training study
Negligent/unintentional discharge rate (qualification failures)	Up to ~61% remediation-failure rate with harder-to-handle pistol designs	As low as ~17% with modern equipment + improved training	300 Negligent Discharges Study
Trigger discipline (finger on trigger before decision to fire)	Common failure point, often unconscious even in trained personnel under stress	Reduced with drilling, but stress can override training	Force Science — preventing unintentional discharges
Decision accuracy under time pressure	Baseline accuracy drops noticeably	Same ~15% accuracy drop applies even to trained shooters under high perceived pressure — expertise reduces but doesn't eliminate this	Stress effects on shooting accuracy